

5.7 Genetic Testing

▼ What is a nuchal scan?

- It is **one of the two main types of ultrasound scan** used for assessing abnormalities in infants

▼ When do pregnant women undergo nuchal scans?

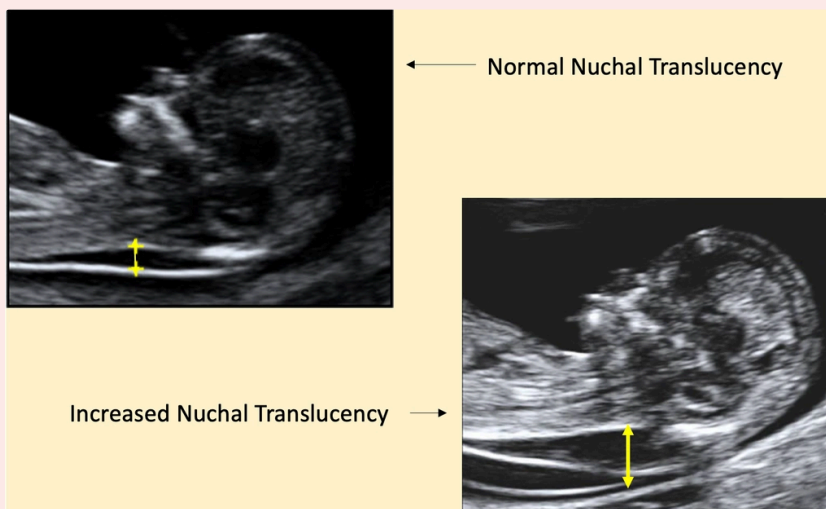
- **10-14 weeks.**

▼ Recall three purposes of nuchal scans.

- To **date the pregnancy** accurately
- To diagnose **multiple pregnancy** (i.e twins, triplets)
- To diagnose **major fetal anomalies**
- To diagnose **early miscarriages**
- To **assess the chance of Down's syndrome** and any **other chromosomal abnormalities**

▼ Describe how a nuchal scan works.

- It uses **Nuchal Translucency (NT)** which measures the **thickness of fluid at the back of a baby's neck**
- If the **thickness is >3mm**, this can indicate **chromosomal abnormalities**



▼ When is an anomaly scan undertaken?

- **20 weeks.**

▼ What is the purpose of an anomaly scan?

- To look for **structural anomalies** specifically in the **heart, brain, spinal cord, face, kidney and abdomen**
- To **measure the lengths of the bones**

▼ What is prenatal testing?

- Testing that is **undertaken during a pregnancy** (before birth) on babies

▼ What is the purpose of prenatal testing?

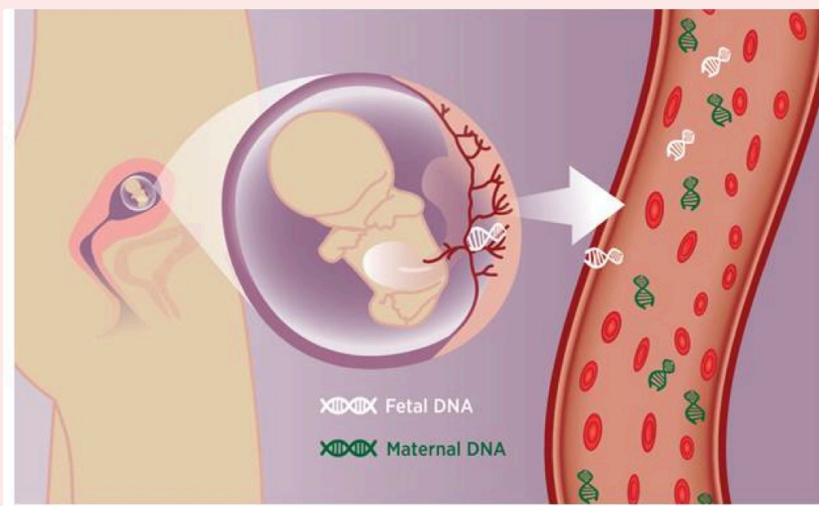
- Following results of the two ultrasound scans, if the combined risk score is over 1/150, then **prenatal testing is used to confirm the abnormalities present**

▼ When can prenatal testing be used?

- Combined risk score of baby is over **1/150**
- Family history of genetic condition
- Parents are carriers of risk genes
- Previous pregnancy has been affected by a condition
- Mainly to check the genotype of the baby for a particular trait they are at risk of

▼ What is non-invasive prenatal diagnosing (NIPD)?

- **Analysis of the DNA fragments** present in the maternal plasma **during pregnancy**
- Different things can be learned from these DNA fragments
- **Most of the DNA** is from the **mother**, **but 10-20% comes from the placenta** which is representative of the fetus
 - the fetal DNA is in the form of cell-free DNA and will contain both the maternal and paternal DNA



▼ When is non-invasive prenatal diagnosing (NIPD) undertaken?

- Around **9 weeks**.

▼ How does NIPD fetal sexing work?

- To **look for the SRY gene** which is **only present on Y chromosomes**
- Hence, if the **fetal DNA has the SRY gene**, we know that **it is a male**

▼ Describe how NIPD is used in genetic testing.

- Because NIPD involves blood from the mother mainly, we are looking for genes from the father
- **Autosomal dominant single gene disorders** inherited from **fathers** are searched for
- **Recessive genes from the father** can also be looked for if the mother and father have different versions of this gene
- **X-linked conditions in males** can be used to determine which X-chromosome they inherited from mother.

▼ Describe how NIPT is used in detecting aneuploidy.

- It is used as a **screening technique, not diagnosis**
- If one is found to have a high **risk of aneuploidy**, invasive prenatal testing is used to confirm it.

▼ Recall two limitations of NIPD/NIPT.

- If there are **twins**, **can't tell whose DNA possesses** which genes.
- **Women with a high BMI** have lower proportions of cell free fetal DNA, and so it is hard to use NIPD for them
- **Invasive testing may be required** to confirm abnormal result

▼ Recall two benefits of NIPD/NIPT.

- **No increased risk of miscarriage**
- **Less expertise required** to perform a blood test than an invasive test

▼ What is purpose of the main invasive tests (CVS/amniocentesis)?

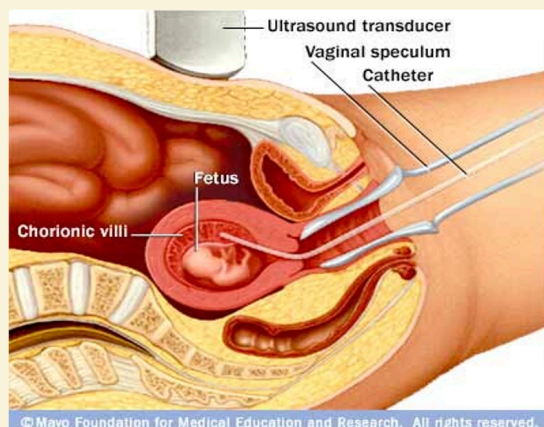
- Invasive testing is offered when there is a **known risk of disorder within the fetus**
- This could be due to family history or NIPD/NIPT results

▼ When is chorionic villus sampling undertaken?

- **12-14 weeks**

▼ How is chorionic villus sampling undertaken?

- A sample of chorionic villi (part of developing placenta) is taken from the woman and analysed
 - This region has the same DNA as fetus



▼ Recall two risks of chorionic villus sampling.

- **0.5% risk of miscarriage**
- **Infection**
- **Rhesus (Rh group) sensitisation**
 - maternal exposure to fetal blood cells

▼ When is amniocentesis undertaken?

- **15 weeks.**
- It **takes sample of amniotic fluid** which contains fetal cells

▼ Recall two risks of amniocentesis.

- 0.5% risk of miscarriage
- Infection
- Rhesus sensitisation (foreign Rh group exposure to woman)

▼ What can be done with the DNA samples obtained from these invasive tests?

- Monogenic disorders can be tested for
- Karyotype analysis
- QF-PCR
 - tests for trisomy 13, 18, 21

▼ What is a comparative genomic hybridization (CGH) array and when is it used?

- A **comparative genomic hybridization (CGH)** array is a test used to determine the abnormalities present in the genome of a fetus
- It is used **when abnormalities are suspected from the 20-week scan.**
- It looks for small/large imbalances in chromosomes (picks up microdeletions and microduplications)

▼ What is a prenatal exome R21 test?

- A test used to examine **where exactly the fetus has had significant anomalies (i.e kidney, heart, brain, skeletal)**
- The **exome is the coding region of the genome**
- It happens at **20+ weeks of pregnancy**
- It is not always conclusive

- ▼ Recall three alternative options that couples who want to have a baby should consider if their baby is at risk of a genetic disorder.
- Conceive naturally, no prenatal testing
 - Conceive naturally, have prenatal testing
 - Pre-implantation genetic diagnosis (**PIGD**)
 - Use of **egg/sperm donors**
 - Adoption